Bihar Engineering University, Patna

B.Tech 8th Semester Examination, 2026

Course: B.Tech
Code: 100815

Subject: Digital Image Processing

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct option / answer the following (Any seven question only):*** **[2 x 7 = 14]**

(a) A grayscale image of size 2048 × 1024 is represented using 12 bits/pixel. The approximate storage requirement is
[✓ (i) 3 MB] (ii) 2 MB
(iii) 1.5 MB (iv) 6 MB

(b) If an image is sampled below the Nyquist sampling rate, the resulting distortion is known as
(i) Quantization error [✓ (ii) Aliasing]
(iii) Blurring (iv) Histogram distortion

(c) For an 8-bit image, the result of applying the transformation s = 255 − r on a pixel of intensity 175 is
(i) 70 (ii) 255
(iii) 175 [✓ (iv) 80]

(d) A histogram concentrated heavily toward the higher gray levels generally indicates
(i) Underexposed image (ii) Low-resolution image
[✓ (iii) Bright image] (iv) Edge-enhanced image

(e) Which filtering technique is most effective for removing impulse noise while preserving edges?
[✓ (i) Arithmetic Mean Filter] (ii) Median Filter
(iii) Gaussian Low-pass Filter (iv) Ideal Low-pass Filter

(f) In the frequency spectrum of an image, abrupt intensity transitions mainly contribute to
[✓ (i) Low-frequency components] (ii) Mid-frequency components
(iii) High-frequency components (iv) Zero-frequency components

(g) The Fourier Transform of a constant image produces
[✓ (i) Uniform frequency response] (ii) Impulse at origin only
(iii) High-pass response (iv) Random frequency distribution

(h) A compression method that removes psychovisual redundancy is generally classified as
(i) Lossy coding (ii) Predictive coding
[✓ (iii) Lossless coding] (iv) Run-length coding

(i) An image compressed from 24 MB to 6 MB has a compression ratio of
(i) 2:1 (ii) 3:1
[✓ (iii) 4:1] (iv) 6:1

(j) If the DFT of an image is multiplied by a high-pass filter mask, the resulting image will mainly emphasize
(i) Smooth regions (ii) Background illumination
(iii) Average intensity [✓ (iv) Edges and fine details]

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Q.2 (a) Explain the fundamental steps involved in Digital Image Processing and discuss the significance of sampling and quantization in image representation. A grayscale image of size 512 × 512 is digitized using 10 bits/pixel. Calculate: [7]

(i) Total number of pixels
(ii) Storage required in bytes and MB
(iii) Number of gray levels represented

(b) Discuss image subtraction and image averaging techniques for noise reduction and background suppression. Two noisy images have pixel intensities 110 and 130 at the same location. Compute: [7]

(i) Averaged pixel intensity
(ii) Difference image value
(iii) Comment on noise reduction effect

Q.3 (a) Define the concepts of adjacency, connectivity, and distance measures between pixels with suitable examples. A binary image contains object pixels at coordinates (2,3), (2,4), (3,4), and (4,5). Determine: [7]

(i) Whether the pixels are 4-connected and 8-connected
(ii) Euclidean distance between pixels (2,3) and (4,5)
(iii) City-block distance between the same pixels

(b) Explain the degradation model used in image restoration and discuss the role of inverse filtering in blur removal. A blurred image is modeled as: [7]

$$g(x,y) = h(x,y) * f(x,y)$$

If $h(x,y) = 0.5$ and observed image intensity $g(x,y)=120$, estimate the restored image intensity using inverse filtering.

Q.4 (a) Explain the properties and significance of the Two-Dimensional Discrete Fourier Transform in image processing. For the sequence f(x) = [2, 4, 6, 8], compute the 1-D DFT coefficients. [7]

(b) What are constrained and unconstrained restoration methods and explain the effects of diagonalization on degradation models. A circulant degradation matrix has eigenvalues [2, 4, 8]. Determine the inverse filter eigenvalues and comment on restoration stability. [7]

Q.5 (a) Describe separable image transforms and compare Fourier Transform with other transforms used in image processing. An image matrix is given as: [1 2] [3 4] Compute the 2-D DFT of the image. [7]

(b) Explain coding redundancy, interpixel redundancy, and psychovisual redundancy in image compression. An image originally requires 8 MB storage. After compression, its size becomes 2 MB. Calculate: [7]

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(i) Compression ratio
(ii) Relative data redundancy

Q.6 (a) Discuss histogram processing techniques and explain how histogram equalization improves image contrast. For an image having gray levels {0,1,2,3} with probabilities {0.1, 0.2, 0.3, 0.4}, determine: [7]

(i) Cumulative distribution function
(ii) Equalized gray level mapping

(b) Describe lossless and lossy image compression techniques with suitable examples. A predictive coding system predicts a pixel value as 120 while actual value is 128. Determine: [7]

(i) Prediction error
(ii) Reconstructed pixel value if error is encoded perfectly

Q.7 (a) Explain smoothing and sharpening filters in spatial domain enhancement with suitable masks. Apply the following averaging filter to the center pixel of the image: [7]

[10 20 30] [40 50 60] [70 80 90]

Using a 3 × 3 mean filter and compute the output pixel value.

(b) Explain transform coding and bit-plane coding methods used in image compression. An 8-bit grayscale image has pixel value 213. Determine its binary representation and identify the most significant bit-plane. [7]

Q.8 (a) Explain low-pass and high-pass filtering in the frequency domain and discuss their applications in image enhancement. An image contains frequency components with amplitudes [12, 20, 5, 2]. If a low-pass filter removes frequencies above threshold 10, determine the remaining components. [7]

(b) Describe the source encoder and channel encoder in image compression systems. A Huffman coding scheme assigns code lengths of 2, 3, 4, and 5 bits to symbols with probabilities of 0.5, 0.25, 0.15, and 0.10, respectively. Compute the average code length. [7]

Q.9 Write short notes on ***any two*** of the following: [7 x 2 = 14]

(a) Challenges in blind image restoration and motion blur compensation

(b) Role of frequency-domain masks in edge enhancement and texture extraction

(c) Applications of transform coding in modern multimedia communication systems

(d) Comparative analysis of spatial-domain and frequency-domain image enhancement techniques

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